Federico

Section 1 — Auth & Users

1. ROLE

Schema

```
ROLE (
role_id INT PRIMARY KEY,
role_name VARCHAR(50) UNIQUE NOT NULL
)
```

Relationship

ROLE (1) —— (N) USER

Participation

USER → total participation

ROLE → partial participation

Meaning: every user must have a role.

Attribute	Role 1	Role 2
role_id	1	2
role_name	HOM	Patient

---

2. USER

Schema

```
USER (
user_id INT PRIMARY KEY,
name VARCHAR(100),
email VARCHAR(100) UNIQUE,
password_hash VARCHAR(255),
role_id INT NOT NULL,
created_at TIMESTAMP,
FOREIGN KEY (role_id) REFERENCES ROLE(role_id)
)
```

Relationships

USER (1) —— (0..1) PATIENT

USER (1) —— (N) PURCHASE_REQUEST

Participation

PATIENT → total

USER → partial

Attribute	User 1	User 2
user_id	101	102
name	Hamiz	Shams
email	hamiz@hosp.com	shams@gmail.com
password_hash	`\$2b\$12\$eJ...`	`\$2b\$12\$kP...`
role_id	1	2
created_at	2026-03-01 10:00:00	2026-03-02 11:30:00

Section 2 — Patients

3. PATIENT

Schema

```
PATIENT (
patient_id INT PRIMARY KEY,
user_id INT NOT NULL,
uhid VARCHAR(20) UNIQUE NOT NULL,
name VARCHAR(100) NOT NULL,
phone VARCHAR(15),
alternate_phone VARCHAR(15),
dob DATE,
gender VARCHAR(10),
blood_group VARCHAR(5),
address TEXT,
emergency_contact_name VARCHAR(100),
emergency_contact_phone VARCHAR(15),
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
FOREIGN KEY (user_id) REFERENCES USER(user_id)
)
```

Relationships

PATIENT (1) —— (N) APPOINTMENT

PATIENT (1) —— (N) ADMISSION

PATIENT (1) —— (N) PATIENT_INSURANCE

PATIENT (1) —— (N) DISCHARGE_SUMMARY

Participation

PATIENT → partial participation

APPOINTMENT → total participation

Attribute	Patient 1	Patient 2
patient_id	201	202
user_id	102	103
uhid	UHID-882100	UHID-994200
name	Hamiz	Shams
phone	+91-9876543210	+91-9123456789
alternate_phone	+91-9876543211	NULL
dob	1998-04-12	1997-08-25
gender	Male	Male
blood_group	O+	A-
address	12 MG Road, Hyderabad, Telangana	45 Beach Road, Visakhapatnam, Andhra Pradesh
emergency_contact_name	Amina Begum	Salma Begum
emergency_contact_phone	+91-9000011111	+91-9888877777
created_at	2026-03-02 11:35:00	2026-03-03 09:15:00

4. PATIENT_INSURANCE

Schema

```
PATIENT_INSURANCE (
insurance_id INT PRIMARY KEY,
patient_id INT NOT NULL,
provider_name VARCHAR(100),
policy_number VARCHAR(50),
member_id VARCHAR(50),
coverage_type VARCHAR(20),
valid_from DATE,
valid_to DATE,
created_at TIMESTAMP,
FOREIGN KEY (patient_id) REFERENCES PATIENT(patient_id)
)
```

Relationships

PATIENT (1) —— (N) PATIENT_INSURANCE

PATIENT_INSURANCE (1) —— (0..N) PATIENT_INSURANCE_DOCUMENTS

Participation

PATIENT_INSURANCE → total

PATIENT → partial

Attribute	Insurance 1	Insurance 2
insurance_id	301	302
patient_id	201	202
provider_name	Niva Bupa	Star Health
policy_number	NB-77210	SH-99102
member_id	M-990	M-112
coverage_type	Full	Basic
valid_from	2025-01-01	2026-01-01
valid_to	2027-12-31	2028-06-30
created_at	2026-03-02	2026-03-03

---

5. PATIENT_INSURANCE_DOCUMENTS

Schema

```
PATIENT_INSURANCE_DOCUMENTS (
doc_id INT PRIMARY KEY,
insurance_id INT NOT NULL,
file_name VARCHAR(255),
file_path VARCHAR(500),
uploaded_at TIMESTAMP,
FOREIGN KEY (insurance_id) REFERENCES PATIENT_INSURANCE(insurance_id)
)
```

Relationship

PATIENT_INSURANCE (1) —— (N) PATIENT_INSURANCE_DOCUMENTS

Participation

PATIENT_INSURANCE_DOCUMENTS → total

PATIENT_INSURANCE → partial

Attribute	Document 1	Document 2
doc_id	1	2

Attribute	Document 1	Document 2
insurance_id	301	302
file_name	policy_copy.pdf	id_card.jpg
file_path	/docs/ins/301/p1.pdf	/docs/ins/302/id.jpg
uploaded_at	2026-03-02 12:00:00	2026-03-03 10:00:00

Section 3 — Doctors

6. DOCTOR

Schema

```
DOCTOR (
doctor_id INT PRIMARY KEY,
name VARCHAR(100),
specialization VARCHAR(100),
phone VARCHAR(15),
email VARCHAR(100)
)
```

Relationship

DOCTOR (1) —— (N) DOCTOR_AVAILABILITY

Participation

DOCTOR_AVAILABILITY → total

DOCTOR → partial

Attribute	Doctor 1	Doctor 2
doctor_id	401	402
name	Dr. Arjun Mehta	Dr. Sneha Reddy
specialization	Cardiology	Neurology
phone	8881112222	8883334444
email	arjun.m@hosp.com	sneha.r@hosp.com

---

7. DOCTOR_AVAILABILITY

Schema

```
DOCTOR_AVAILABILITY (
availability_id INT PRIMARY KEY,
doctor_id INT NOT NULL,
available_date DATE,
start_time TIME,
end_time TIME,
status VARCHAR(20),
FOREIGN KEY (doctor_id) REFERENCES DOCTOR(doctor_id)
)
```

Relationships

DOCTOR (1) —— (N) DOCTOR_AVAILABILITY

DOCTOR_AVAILABILITY (1) —— (0..N) APPOINTMENT

Attribute	Availability 1	Availability 2
availability_id	501	502

Attribute	Availability 1	Availability 2
doctor_id	401	402
available_date	2026-03-15	2026-03-15
start_time	09:00:00	14:00:00
end_time	12:00:00	17:00:00
status	Available	Available

---

8. APPOINTMENT

Schema

```
APPOINTMENT (
appointment_id INT PRIMARY KEY,
patient_id INT NOT NULL,
availability_id INT NOT NULL,
scheduled_datetime TIMESTAMP NOT NULL,
visit_type VARCHAR(20)
CHECK (visit_type IN ('OPD','EMERGENCY','FOLLOWUP')),
status VARCHAR(20)
CHECK (status IN ('PENDING','CONFIRMED','CANCELLED','COMPLETED')),
created_by INT,
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
FOREIGN KEY (patient_id) REFERENCES PATIENT(patient_id),
FOREIGN KEY (availability_id) REFERENCES DOCTOR_AVAILABILITY(availability_id),
FOREIGN KEY (created_by) REFERENCES USER(user_id)
)
```

Relationships

PATIENT (1) —— (N) APPOINTMENT

DOCTOR_AVAILABILITY (1) —— (N) APPOINTMENT

APPOINTMENT (1) —— (0..1) ADMISSION

Participation

APPOINTMENT → total

PATIENT → partial

Attribute	Appointment 1	Appointment 2
appointment_id	601	602
patient_id	201	202
availability_id	501	502
scheduled_datetime	2026-03-15 09:30:00	2026-03-15 14:15:00
visit_type	OPD	FOLLOWUP
status	CONFIRMED	COMPLETED
created_by	101	101
created_at	2026-03-10	2026-03-11

Section 4 — Wards

9. WARD

Schema

```
WARD (
ward_id INT PRIMARY KEY,
ward_name VARCHAR(50) UNIQUE NOT NULL,
total_beds INT NOT NULL,
description TEXT
)
```

Relationship

WARD (1) —— (N) BED

Participation

BED → total

WARD → partial

Attribute	Ward 1	Ward 2
ward_id	1	2
ward_name	General Ward A	ICU - 01
total_beds	20	10
description	North Wing Floor 1	Critical Care Unit

---

10. BED

Schema

```
BED (
bed_id INT PRIMARY KEY,
ward_id INT NOT NULL,
bed_number VARCHAR(10),
status VARCHAR(20),
FOREIGN KEY (ward_id) REFERENCES WARD(ward_id)
)
```

Relationships

WARD (1) —— (N) BED

BED (1) —— (N) ADMISSION

Attribute	Bed 1	Bed 2
bed_id	11	22
ward_id	1	2
bed_number	G-101	ICU-05
status	OCCUPIED	AVAILABLE

---

11. ADMISSION

Schema

```
ADMISSION (
admission_id INT PRIMARY KEY,
appointment_id INT NOT NULL,
patient_id INT NOT NULL,
bed_id INT NOT NULL,
admit_time TIMESTAMP,
discharge_time TIMESTAMP,
status VARCHAR(20),
FOREIGN KEY (appointment_id) REFERENCES APPOINTMENT(appointment_id),
```

```
FOREIGN KEY (patient_id) REFERENCES PATIENT(patient_id),
FOREIGN KEY (bed_id) REFERENCES BED(bed_id)
)
```

Relationships

APPOINTMENT (1) —— (0..1) ADMISSION

PATIENT (1) —— (N) ADMISSION

BED (1) —— (N) ADMISSION

ADMISSION (1) —— (1) LEDGER

ADMISSION (1) —— (0..1) INSURANCE

ADMISSION (1) —— (1) DISCHARGE_SUMMARY

Participation

ADMISSION → total

APPOINTMENT → partial

Attribute	Admission 1	Admission 2
admission_id	701	702
appointment_id	601	602
patient_id	201	202
bed_id	11	22
admit_time	2026-03-15 10:30:00	2026-03-16 08:00:00
discharge_time	2026-03-18 10:00:00	NULL
status	DISCHARGED	ADMITTED

---

12. DISCHARGE_SUMMARY

Schema

```
DISCHARGE_SUMMARY (
summary_id INT PRIMARY KEY,
admission_id INT UNIQUE,
patient_id INT,
discharge_notes TEXT,
final_amount DECIMAL(10,2),
generated_at TIMESTAMP,
file_path VARCHAR(255),
FOREIGN KEY (admission_id) REFERENCES ADMISSION(admission_id),
FOREIGN KEY (patient_id) REFERENCES PATIENT(patient_id)
)
```

Relationships

ADMISSION (1) —— (1) DISCHARGE_SUMMARY

PATIENT (1) —— (N) DISCHARGE_SUMMARY

Attribute	Summary 1	Summary 2
summary_id	1	2
admission_id	701	702
patient_id	201	202
discharge_notes	Recovered well.	Followup in 7 days.
final_amount	15500.00	8200.00
generated_at	2026-03-18	2026-03-19
file_path	/path/ds_201.pdf	/path/ds_202.pdf

Section 5 — Billing

13. SERVICE

Schema

```
SERVICE (
service_id INT PRIMARY KEY,
service_name VARCHAR(100),
base_cost DECIMAL(10,2)
)
```

Relationships

SERVICE (1) —— (N) LEDGER_ENTRY

SERVICE (1) —— (N) INVENTORY_ITEM

Attribute	Service 1	Service 2
service_id	1	2
service_name	Consultation Fee	Room Rent
base_cost	500.00	5000.00

---

14. LEDGER

Schema

```
LEDGER (
ledger_id INT PRIMARY KEY,
admission_id INT UNIQUE,
status VARCHAR(20),
created_at TIMESTAMP,
FOREIGN KEY (admission_id) REFERENCES ADMISSION(admission_id)
)
```

Relationships

ADMISSION (1) —— (1) LEDGER

LEDGER (1) —— (N) LEDGER_ENTRY

LEDGER (1) —— (N) PAYMENT

Attribute	Ledger 1	Ledger 2
ledger_id	801	802
admission_id	701	702
status	CLOSED	OPEN
created_at	2026-03-15	2026-03-16

---

15. LEDGER_ENTRY (Weak Entity)

Schema

```
LEDGER_ENTRY (
entry_id INT,
ledger_id INT NOT NULL,
service_id INT,
quantity INT,
unit_price DECIMAL(10,2),
```

```
amount DECIMAL(10,2),
entry_time TIMESTAMP,
PRIMARY KEY (entry_id, ledger_id),
FOREIGN KEY (ledger_id) REFERENCES LEDGER(ledger_id),
FOREIGN KEY (service_id) REFERENCES SERVICE(service_id)
)
```

Relationships

LEDGER (1) —— (N) LEDGER_ENTRY

SERVICE (1) —— (N) LEDGER_ENTRY

Participation

LEDGER_ENTRY → total

LEDGER → partial

Owner entity: LEDGER

Attribute	Entry 1	Entry 2
entry_id	1	2
ledger_id	801	801
service_id	1	2
quantity	1	3
unit_price	500.00	5000.00
amount	500.00	15000.00
entry_time	2026-03-15	2026-03-15

---

16. INSURANCE

Schema

```
INSURANCE (
insurance_id INT PRIMARY KEY,
admission_id INT UNIQUE NOT NULL,
patient_insurance_id INT NOT NULL,
coverage_percentage DECIMAL(5,2),
claim_status VARCHAR(20)
CHECK (claim_status IN ('PENDING','APPROVED','REJECTED')),
FOREIGN KEY (admission_id) REFERENCES ADMISSION(admission_id),
FOREIGN KEY (patient_insurance_id) REFERENCES PATIENT_INSURANCE(insurance_id)
)
```

Relationship

ADMISSION (1) —— (0..1) INSURANCE

Attribute	Claim 1	Claim 2
insurance_id	1	2
admission_id	701	702
patient_insurance_id	301	302
coverage_percentage	80.00	50.00
claim_status	APPROVED	PENDING

---

17. PAYMENT

Schema

```
PAYMENT (
payment_id INT PRIMARY KEY,
ledger_id INT NOT NULL,
amount_paid DECIMAL(10,2),
payment_mode VARCHAR(20),
payment_time TIMESTAMP,
FOREIGN KEY (ledger_id) REFERENCES LEDGER(ledger_id)
)
```

Relationship

LEDGER (1) —— (N) PAYMENT

Attribute	Payment 1	Payment 2
payment_id	901	902
ledger_id	801	802
amount_paid	3100.00	1000.00
payment_mode	CASH	UPI
payment_time	2026-03-18	2026-03-16

Section 6 — Inventory

18. INVENTORY_ITEM

Schema

```
INVENTORY_ITEM (
item_id INT PRIMARY KEY,
item_name VARCHAR(100),
category VARCHAR(50),
stock_quantity INT,
reorder_level INT,
service_id INT,
FOREIGN KEY (service_id) REFERENCES SERVICE(service_id)
)
```

Relationships

INVENTORY_ITEM (1) —— (N) PURCHASE_REQUEST

SERVICE (1) —— (N) INVENTORY_ITEM

Attribute	Item 1	Item 2
item_id	10	20
item_name	Syringe 5ml	Paracetamol
category	Consumable	Medicine
stock_quantity	500	1200
reorder_level	100	200
service_id	NULL	1

---

19. PURCHASE_REQUEST

Schema

```
PURCHASE_REQUEST (
request_id INT PRIMARY KEY,
```

```
item_id INT,
quantity_requested INT,
status VARCHAR(20),
requested_by INT,
requested_at TIMESTAMP,
FOREIGN KEY (item_id) REFERENCES INVENTORY_ITEM(item_id),
FOREIGN KEY (requested_by) REFERENCES USER(user_id)
)
```

Relationships

INVENTORY_ITEM (1) —— (N) PURCHASE_REQUEST

USER (1) —— (N) PURCHASE_REQUEST

Attribute	Request 1	Request 2
request_id	1001	1002
item_id	10	20
quantity_requested	1000	500
status	APPROVED	PENDING
requested_by	101	101
requested_at	2026-03-05	2026-03-06